David Alejandro Torres Torres

Mechatronics Engineer | Swarm Robotics & Multi-Robot Systems Researcher

CONTACT

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SKILLS

Core Research Areas

- Swarm robotics
- Collective intelligence
- Multi-robot systems
- Autonomous navigation
- Robotic manipulation
- Industrial robotics

Robotics & Autonomy Stack

- ROS 1 and ROS 2
- Gazebo Simulator
- ARGoS (large-scale swarm simulation)
- URDF / Xacro robot modeling
- TF2 and RViz
- Nav2 (planning, controllers, actions)
- Simulation-to-control pipelines

Programming & Control

- C++, Python
- MATLAB, Simulink
- Javascript
- Flutter
- State machines
- Distributed control systems
- Closed-loop control systems
- Signal scaling and normalization

Industrial Automation

- PLC-based control systems
- Tia Portal
- Logo! Soft Comfort
- HMI–PLC integration (Siemens, Flexem)
- Industrial safety systems
- Variable-frequency drives (INVT)
- Power factor correction systems
- Motor control
- Sensor integration (pressure sensors and switches)

ABOUT

I am a mechatronics engineer and prospective PhD researcher focused on human-centered robotics, collective robotic systems, and interdisciplinary design for socially impactful applications.

My work integrates swarm robotics, robotic manipulation, and decentralized control with hands-on mechatronic system design, emphasizing safety-critical and socially engaged domains such as construction robotics, assistive technologies, and STEM education.

I am particularly interested in bio-inspired and collective robotic systems, human–robot interaction, and the development of intelligent learning technologies that connect robotics, design, and public engagement.

PROFESSIONAL EXPERIENCE

Research Assistant — Swarm Robotics & Collective Behavior 2024 – Present (Part-time Research)
Collective Robotics and Artificial intelligence Laboratory (CoRAL), ESPOL

- Conducted research under the supervision of Dr. Christian Tutiven Galvez (Head of CoRAL at Escuela Superior Politécnica del Litoral (ESPOL), Ecuador) who served as my undergraduate thesis advisor and primary academic mentor.
- Worked in collaboration with Prof. David Garzón Ramos (Assistant Professor and Ad Astra Fellow, University College Dublin (UCD), Ireland School of Mechanical & Materials Engineering) on swarm robotics and collective construction research.
- Conducted research on decentralized multi-robot systems for swarm robotics and collective behavior in industrial and construction-oriented environments.
- Designed and implemented cooperative robot behaviors, task allocation mechanisms, and distributed task execution pipelines using ROS-based simulation platforms.
- Developed multi-fidelity simulation infrastructures combining ARGoS for large-scale swarm experiments and ROS/Gazebo for physically grounded embodiment and interaction.
- Implemented ROS 1/2 nodes in C++ and Python for swarm behaviors, navigation, and task execution.
- Integrated perception pipelines, state machines, and control logic into a full simulation-to-control architecture.
- Built the foundational swarm robotics simulation platform at ESPOL, including ROS 1-based simulators, URDF robot models, Gazebo plugins, C++ swarm behaviors, and ARGoS-based large-scale swarm experiments, which later evolved into the CoRAL research infrastructure.

Research Assistant — Autonomous Robotics 2023 – 2024 (Remote)
Robotics and Intelligent Systems Engineering (RISE) Lab, Sungkyunkwan University (SKKU), South Korea

- Collaborated with Prof. Nabih Pico, Research Professor at Sungkyunkwan University (SKKU), South Korea, on autonomous navigation, perception, and intelligent robotic system design.
- Developed autonomous navigation pipelines for mobile robots using ROS 2 and Gazebo in structured driving environments.

Mechatronics & System Design

- Actuator design
- Mechanism design
- Cyber-physical system integration
- Sensor–actuator integration
- Electrical control panel design
- System architecture
- Embedded systems (ESP32, Arduino)
- Real-time device control

CAD & Engineering Tools

- Autodesk Inventor
- SolidWorks
- Fusion 360
- Proteus
- EasyEda
- Blender

Leadership & Mentoring

- Team leadership
- Student mentoring
- Technical training
- Academic outreach

PROJECTS

Robotic Surface Printing System for Mobile Robots (2024).
Designed and integrated a modular surface-printing attachment for a TurtleBot3 mobile robot to enable automated printing on planar surfaces.

Quadruped Leg Controller Test Platform (2023).
Developed a dynamic test platform to evaluate motor and joint behavior for quadruped robot legs using impedance-based control. Implemented and validated the system using MATLAB Simscape Multibody.

SmartHatch — IoT-Based Smart Incubator System (2023).
Designed and implemented a smart incubator system with real-time environmental monitoring and control.

3D-Printed Upper-Limb Prosthesis (2024).
Designed and fabricated a low-cost 3D-printed upper-limb prosthesis for a patient who lost an arm in a workplace accident. Delivered as part of a community professional service initiative.

ACADEMIC REFERENCES

Dr. Christian Tutiven Galvez
Head of the Collective Robotics and Artificial Intelligence Laboratory (CoRAL)
Professor, Escuela Superior Politécnica del Litoral (ESPOL), Ecuador
Email: cjtutiven@espol.edu.ec

- Designed and implemented Nav2-based planning and control architectures, including planners, controllers, and action servers.
- Built simulation scenarios and benchmarking environments to evaluate AI-based and classical navigation controllers.
- Conducted comparative evaluations of learning-based and traditional control strategies based on navigation performance, stability, and response behavior.

Automation Engineer — Industrial Systems 2023 – 2025 (Hybrid, Part-time)
WG Selcon, Ecuador

- Designed and deployed PLC-based industrial automation systems for safety-critical material handling, compression, and power conditioning infrastructure operating in 24/7 industrial environments.
- Led end-to-end development of industrial control systems, including system architecture, electrical design, control logic development, hardware integration, and on-site commissioning.
- Engineered electrical control panels for compressors, hydraulic rolling machinery, and power factor correction systems, including component selection, wiring schematics, and panel assembly.
- Developed PLC control software for motor control, sequencing, interlocks, and distributed actuation using standardized industrial architectures, primarily based on Siemens PLC platforms.
- Implemented signal scaling, filtering, and normalization pipelines for industrial sensors (pressure sensors and switches) to enable robust closed-loop monitoring and control.
- Integrated motor starters, compressor control units, and inductive load compensation systems into unified automation pipelines, including systems using INVT variable-frequency drives.
- Developed integrated HMI-PLC solutions using Siemens and Flexem platforms to provide intuitive operator interfaces and support safe, reliable, and user-centered operation of industrial systems.
- Implemented safety logic, fault detection, and operational interlocks to protect equipment and personnel in high-power electrical and hydraulic systems.
- Commissioned and deployed live automation systems into operational plants, including debugging, tuning, and on-site validation under real production constraints.
- Collaborated with mechanical and process engineering teams to retrofit legacy industrial machinery with modern PLC-based control systems.

PUBLICATIONS & INTELLECTUAL PROPERTY

Modular Robotic Platform and Manipulation System for Swarm Construction Robots 2025-Present
Industrial secret (filing stage).
Co-inventor of a modular robotic actuator designed for swarm robotic platforms capable of collective manipulation and construction tasks.
Primary contributions included the mechanical and electromechanical design of the actuator, CAD modeling and drawings of the robot base and actuator assembly, schematic design of actuator electronics, and integration of actuator control with the mobile robotic platform.
Contributed to the system-level design of the base robot and supported the perception, sensing, and calibration subsystems.

LEADERSHIP & SERVICE

Founder Member and Former Vice President — Artificial Intelligence and Robotics Society club (AIRoS) 2024 – 2025
Escuela Superior Politécnica del Litoral (ESPOL), Ecuador

- Co-founded AIRoS, a student-led robotics and artificial intelligence research and outreach group focused on autonomous navigation, perception, and ROS-based robotic systems.
- Recruited and mentored undergraduate students, introducing them to robotics

Prof. David Garzón Ramos
Assistant Professor and Ad Astra Fellow
School of Mechanical & Materials Engineering
University College Dublin (UCD), Ireland
Email: david.garzonramos@ucd.ie
Role: Research collaborator in swarm robotics and collective construction

Prof. Nabih Pico
Research Professor
Robotics and Intelligent Systems Engineering (RISE) Lab
Sungkyunkwan University (SKKU), South Korea
Email: npico@g.skku.edu

fundamentals, ROS, and the development of autonomous navigation systems.

- Led technical training activities and project-based learning initiatives in mobile robotics, perception, and control.
- Coordinated academic talks and outreach events across universities and high schools, featuring international PhD students presenting current research in robotics and AI.
- Represented ESPOL and led a student team whose autonomous racing platform won first place, while a second AIRoS team won second place at Roboracer Ecuador.
- Helped establish a research-oriented culture within the club, contributing to the production of 3 conference papers and 2 journal articles by student members in one year.

HONORS & AWARDS

First Place — Roboracer Ecuador (Autonomous Driving Competition) 2024
Mentored and led the winning autonomous racing team representing ESPOL through AIRoS.

EDUCATION

B.Sc. in Mechatronics Engineering 2025
Escuela Superior Politécnica del Litoral (ESPOL), Ecuador

- Thesis: *Swarm Robotic Manipulation and Collective Transport for Construction-Oriented Tasks*

High School Diploma — Unified Science Track 2019
Unidad Educativa Javier, Ecuador

English Proficiency
Test of English as a Foreign Language (TOEFL iBT) 2025
Score: 86

Additional Training

Cátedra Empresarial de Innovación (ESPOL–Holcim) 2025
40-hour academic–industry training program on innovation management, sustainability, and organizational transformation.